Electronic coarse-graining for accurate and scalable modeling of organic semiconductors

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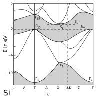
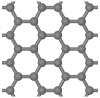
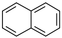
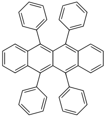
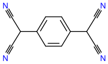
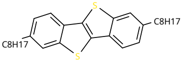
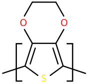
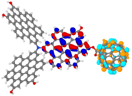
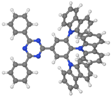
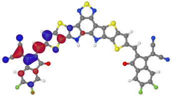
Outline

1) Motivation, 2) Methodology, 3) Applications

Motivation: Accurate modeling of organic semiconductors

$\Longrightarrow$ Coarse-graining of molecular and electronic degrees of freedom

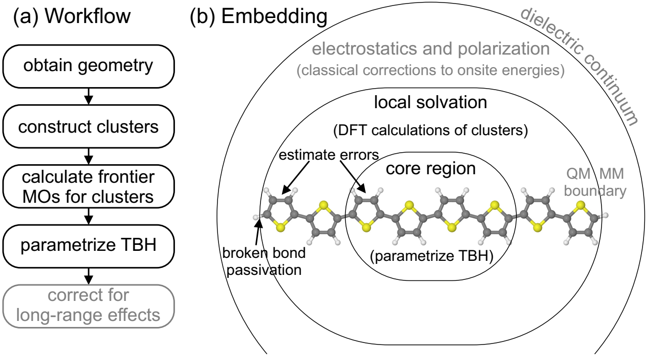
Electronic coarse-graining — reducing size of electronic basis w/o loss of accuracy

class	basis	examples
✓ Broadband semiconductors ("free" e/h)	valence orbitals	Si , graphene , black P, GaAs
✓ Small-molecule organic semiconductors (intermolecular hopping)	HOMO /LUMO	    naphthalene, rubrene, TCNQ, C8-BTBT
?? Other organic semiconductors (intra- and intermolecular dynamics)	fragment HOMO /LUMO	    polymers, large molecules, MOF, OLED, OPV

Goal is to model complex organic semiconductors at level of small-molecule ones: 1) black-box, 2) scalable, 3) compatible technology, 4) controllable accuracy

Methodology of electronic coarse-graining: Main ideas

- We need only frontier molecular orbitals
- Large enough clusters approximate infinite solid



TBH = Tight Binding Hamiltonian

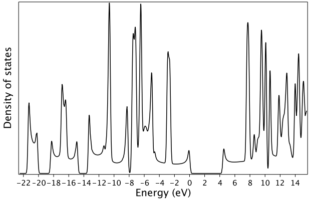
Technical details are available at https://cmsos.github.io/escp

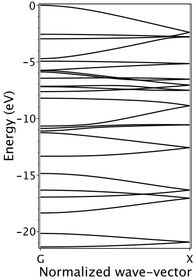
... will be illustrated with two examples: polythiophene and graphene

Electronic coarse-graining: example of polythiophene

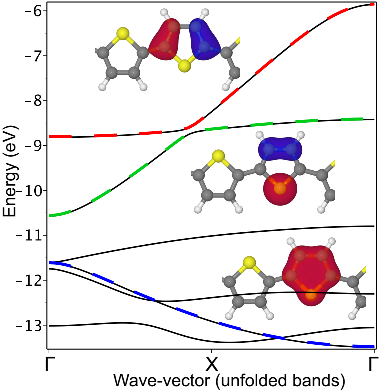
(valence band of single planar polythiophene chain, min basis = 2 LMOs per monomer)

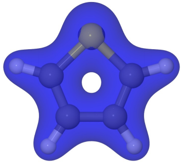
bands in all-electron basis



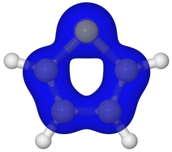


bands in coarse-grained basis





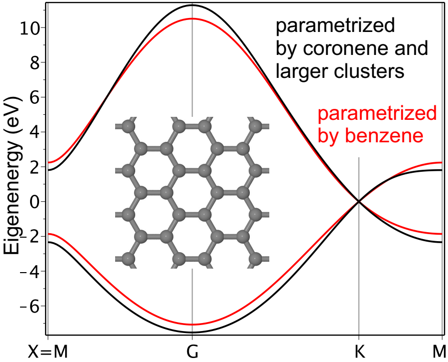
all-electron density

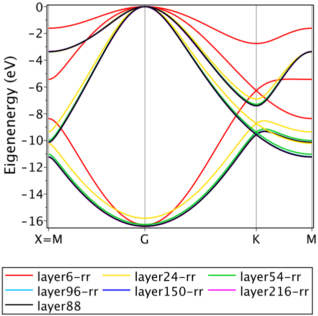


CG density (2 MOs)

The proposed approach solves electronic structure problem for any π-conjugated system that can be fragmentized into π-closed-shell monomers

Electronic coarse-graining: example of graphene





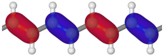
The same approach can be applied to other semiconductors

e.g. to calculate electronic structure of infinite systems with methods available only for finite systems

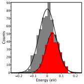
Applications

(where conventional approaches produce substantial systematic errors)

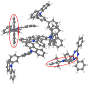
- Electronic structure of crystalline polymers



- Density of electronic states in organic semiconductors



- Guest-to-guest tunneling in host-guest systems



- Electronic transport through A-D-A molecules for solar cells

  J Chem Phys 159, 024107 (2023) DOI

- Charge storage mechanism in metal-organic polymers

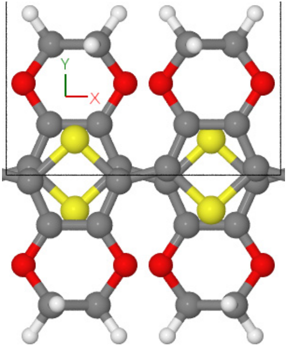
  Chem Sci 13, 8161 (2022) DOI

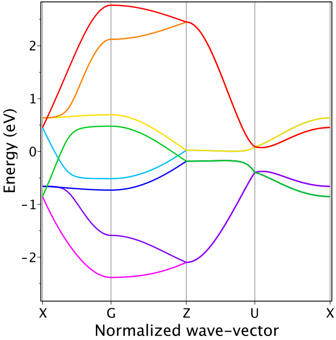
Electronic structure of crystalline polymers

Challenges

- Combination of strong intrachain and weak interchain couplings
- Unit cell and supercell are too large for brute force methods
- No experimental crystal structure JPCC 122, 9141 (2018) DOI; JCTC 19, 8481 (2023) DOI

$\Longrightarrow$ Little is known about electronic structure of crystalline polymers

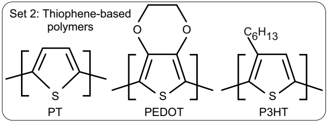


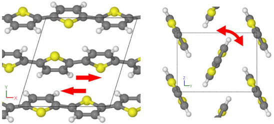


Electronic coarse-graining allows for calculation of electronic structure of polymers at the level of accurate density functionals

Why PEDOT is one of best polymers for charge transport?

(Here we compare only thiophene-based polymers: PolyThiophene, PEDOT, P3HT)





- How large intermolecular electronic couplings can be?
  250 meV in perfectly aligned π-stacks (max-symmetry π-stack of PPP)

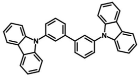
- What prevents perfect alignment?
  Flat potential energy surface wrt to tilts and shifts
  (0 to max coupling upon shift by 1 bondlength due to BLA pattern)

- What is the "best" thiophene-based polymer?
  2D PEDOT (e.g. PEDOT:PSS) –
  (1) well-aligned to maximize intermolecular couplings
  (2) small thermal fluctuations of these couplings

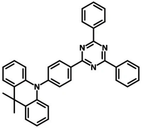
How to correctly compare simulated electronic DOS?

(Critically important for heterogeneous systems such as host-guest systems)

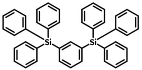
Problem: Often DOS is approximated by HOMO- or LUMO-DOS



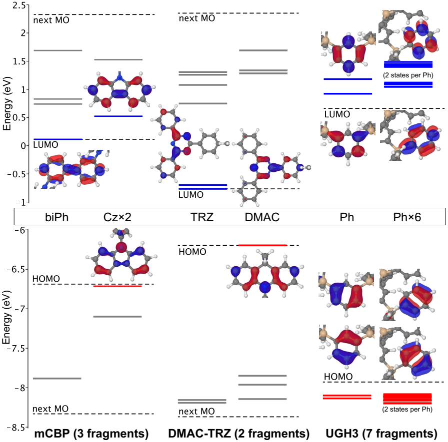
mCBP (host)

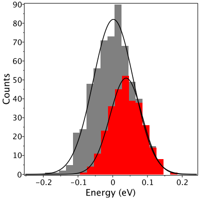


DMAC-TRz (guest)

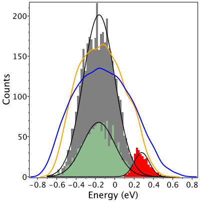


UGH3 (host)





holes in mCBP



holes in UGH3

Right panel: in red – HOMO-DOS (biased), in gray – full DOS (correct)

Compare complete electronic DOS for electronically different systems

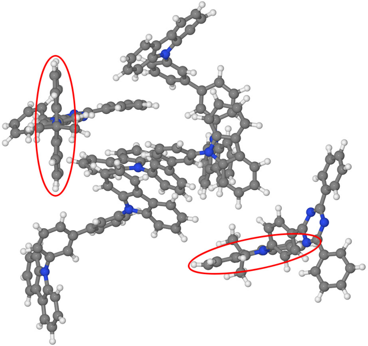
(e.g. UGH vs other OLED host and guest molecules)

Is guest-to-guest tunneling efficient in host-guest systems?

How to accurately calculate tunneling couplings (“superexchange”)?

Challenges leading to loss of accuracy

- Effects of environment (correct SCF)
- Multiple pathways
- Electronic localization



Contact distance is 12 Å,
3 bridging (“tunnel”) molecules,
coupling is 0.13 meV

Couplings are too small, see talk CPP57.2 by Naomi Kinaret

Summary

Electronic Coarse-Graining is a powerful tool (sometimes the only one) for calculation and analysis of electronic structure of complex systems:

- Anything that can be monomerized (polymers, COFs, MOFs)
- Solids and clusters of large molecules
- Defects in solids
- Infinite systems from finite-size calculations

Project web-page

- Electronic Coarse-Graining (including copy of this presentation)
  https://zhugayevych.me/research/ECG
- Computational Materials Science of Organic Semiconductors
  https://cmsos.github.io